

1. Administrative & Study Identification

Full Study Title A Randomized Double-blind, Placebo-controlled, Multicenter Trial Assessing the Impact of Lipoprotein(a) Lowering With Pelacarsen (TQJ230) on the Progression of Calcific Aortic Valve Stenosis [Lp(a)FRONTIERS CAVS] **Short Title/Acronym** Lp(a)FRONTIERS CAVS **Protocol Number** CTQJ230A12203 **NCT Number** NCT05646381 **Sponsor Name** Novartis Pharmaceuticals / Novartis Pharma AG **Investigational Product** Pelacarsen (TQJ230) 80mg **Phase of Development** Phase 2 **Medical Monitor** Novartis Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the efficacy, safety, and tolerability of pelacarsen (TQJ230) administered subcutaneously once monthly compared to placebo in slowing the progression of calcific aortic valve stenosis (CAVS) over a 3-year period. **Secondary Objectives:** To evaluate the change in Lipoprotein(a) [Lp(a)] levels from baseline and assess the overall safety and tolerability profile of the study drug.

2.2 Background & Rationale Calcific aortic valve stenosis (CAVS) is the most frequent form of valvular heart disease in developed countries. Currently, there is no approved pharmacological therapy that slows the progression of CAVS. Elevated Lp(a) levels (>175 nmol/L or ~70 mg/dL) are strongly and independently associated with accelerated progression of aortic valve calcification and hemodynamic deterioration. Pelacarsen is an antisense oligonucleotide designed to selectively lower the body's Lp(a) production. This study aims to determine whether robust Lp(a) reduction can translate into a disease-modifying clinical benefit by slowing valve calcification progression.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1 ratio)

3.2 Planned Enrollment & Duration **Target \$N\$:** Approximately 502 participants **Number of Sites:** Multicenter (hospital research sites globally) **Estimated Study Start:** 07-Mar-2024 **Estimated Primary Completion:** 12-Mar-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female ≥ 50 to <80 years of age at Screening. 2. Lp(a) ≥ 175 nmol/L at the screening visit, measured at the Central laboratory. 3. Mild or moderate calcific aortic valve stenosis. 4. At the randomization visit, the participant must be optimally treated for existing cardiovascular (CV) risk factors.

4.2 Exclusion Criteria 1. Severe calcific aortic valve stenosis. 2. Uncontrolled hypertension. 3. History of malignancy of any organ system. 4. History of hemorrhagic stroke or other major bleeding. 5. Platelet count $\leq$ Lower Limit of Normal (LLN). 6. Active liver disease or hepatic dysfunction, significant kidney disease, or pregnant/nursing women.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pelacarsen (TQJ230) 80 mg or matching placebo administered once monthly. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 3 years (36 months).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for existing cardiovascular risk factors (e.g., statins, anti-hypertensives). **Prohibited:** Other investigational therapies or commercial agents specifically intended to lower Lp(a) outside of the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Medical History, Lp(a) Labs, Baseline Echo/CCT
2	0 to 3	Randomization, First SC Injection, Health Check
3	3 to 6	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
4	6 to 9	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
5	9 to 12	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
6	12 to 15	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
7	15 to 18	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
8	18 to 21	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
9	21 to 24	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
10	24 to 27	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
11	27 to 30	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
12	30 to 33	Interim SC Injection, Safety Labs, Echo, ECG, Questionnaires
13	33 to 36	Final Echo, CCT Scan, Exit Interview, IP Accountability

(Note: Participants visit hospital research sites every 3 months for health checks, blood/urine donation, and cardiac monitoring)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 3-year rate of progression of calcific aortic valve stenosis. **Measurement Method:** Change in peak aortic jet velocity (Vpeak) assessed by echocardiography, and change in aortic valve calcium (AVC) score quantified by cardiac computed tomography (CCT).

7.2 Secondary & Exploratory Endpoints 1. Change in Lp(a) levels from baseline over the 3-year period. 2. Safety and tolerability assessments, including incidence of adverse events and serious adverse events (SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection via patient questionnaires, physical exams, and lab assessments every 3 months. **Serious Adverse Events (SAEs):** Typically reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent external committee utilized to review unblinded safety and efficacy data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analysis; Safety Set for all adverse event evaluations. **Sample Size Justification:** $N \approx 502$ appropriately powered ($\alpha = 0.05$, $Power = 80\%$) to detect a statistically significant reduction in the progression of Vpeak and AVC scores between the pelacarsen and placebo arms over 3 years. **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) and/or Multiple Imputation for continuous longitudinal imaging data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to ensure protocol adherence and patient safety. **Audit Trail:** Robust eCRF locking, tracking of central imaging core lab reading (Echo and CCT), and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: CTQJ230A12203 **Registry Number:** NCT05646381 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Lp(a)FRONTIERS CAVS **Official Regulatory Title:** A Randomized Double-blind, Placebo-controlled, Multicenter Trial Assessing the Impact of Lipoprotein(a) Lowering With Pelacarsen (TQJ230) on the Progression of Calcific Aortic Valve Stenosis

12. Clinical Framework (CAVS & Lp(a) Specific)

Primary Condition: Calcific Aortic Valve Stenosis (CAVS)
Diagnosis Threshold: Lp(a) ≥ 175 nmol/L (70 mg/dL)
Medical Methodology: Lipoprotein(a) lowering therapy
Optimization Target: Delaying the progressive remodeling, thickening, and calcification of the aortic valve

13. Study Procedures & Methodology

Management Pathway: Monthly SC injections in conjunction with optimized standard cardiovascular care.
Education Protocol (HCP): Site training on specialized imaging protocols (Echo/CCT) and SC injection administration.
Education Protocol (Patient): General health education, side effect tracking, and potential self-administration training for SC injections.
Quality Audit Mechanism: Centralized core laboratory reading for all echocardiography and CCT scans to ensure blinded, unbiased progression tracking.

14. Observation & Follow-Up Schedule

Total Duration: 3 Years (36 months)
Onsite Visit Cadence: Every 3 months (Quarterly)
Remote Monitoring Frequency: Intermittent remote check-ins between quarterly visits as needed.
Checklist Review Interval: Every 3 months (Quality of life, general health, and side-effect questionnaires)

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Statistician	[Name]	_____	[DD-MMM-YYYY]						